Demonstrating CatDB: LLM-based Generation of Data-centric ML Pipelines

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Abstract

AutoML systems automate finding Machine Learning (ML) pipelines but struggle to scale with large datasets due to time-consuming data analysis and complex hyper-parameter search spaces. LLMs (Large Language Models) offer flexibility and scalability for code generation with strong generalization across coding tasks. However, generating data-centric ML pipeline scripts is more challenging, as it requires complex reasoning to align the needs of a dataset with coding tasks, such as data cleaning or feature transformation. Thus, LLMs struggle to generate effective and efficient ML pipelines. This demo paper presents CatDB, which overcomes these challenges by dynamically generating dataset-specific instructions to guide LLMs in generating effective pipelines. CatDB profiles datasets to extract metadata, including refined data catalog information and statistics, and then uses this metadata to break down pipeline generation into instructions of tasks such as data cleaning, transformation, and model training, tailored to specifics of the dataset at hand. This process enables CatDB to leverage LLM coding capabilities more effectively. Our evaluation shows CatDB outperforms existing LLM-based and AutoML systems with up to orders of magnitude faster runtime on large datasets. The audience will experience CatDB's capabilities with commercial and open-source LLMs, using a variety of real datasets, as shown in our demo video and Colab notebook.

CCS Concepts

• **Information systems** → **Deduplication**; **Data cleaning**; **Data analytics**; **Online analytical processing**; • **Computing methodologies** → *Machine learning*.

Keywords

Data-centric ML Pipelines; Data Catalogs; LLM Code Generation

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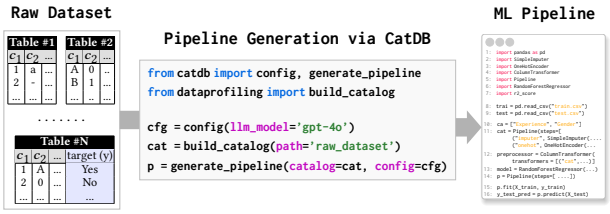


Figure 1: Data-centric ML Pipeline Generation via CatDB.

1 Introduction

Data-centric machine learning (ML) pipelines build on traditional ML pipelines by emphasizing key data preparation steps, such as cleaning, transformation, and feature engineering. These stages are crucial because the dataset quality often has a greater impact on model accuracy than the choice of algorithm. For example, a data scientist working with a tabular dataset aims to create a pipeline that optimizes performance for tasks like classification or regression. This process involves preparing the dataset through dedicated primitives—such as handling missing values or scaling numerical features—to enhance model performance. Data-centric pipelines are inherently exploratory and require iterative experimentation and a deep understanding of the dataset's quality. Hence, designing high-quality pipelines is computationally intensive and challenging, especially for domain-specific or large datasets. Human-in-the-loop approaches heavily depend on expert data scientists to refine the pipeline, making the process time-consuming and difficult to scale.

AutoML systems automate finding good ML pipelines through model selection and hyper-parameter tuning. However, these systems struggle to scale with large datasets due to time-consuming training and complex hyper-parameter search spaces. In contrast, large language models (LLMs), such as GPT-4 and Gemini-2, offer flexibility and scalability for code generation with strong generalization across coding tasks. Unlike code generation, generating ML pipelines requires in-depth reasoning to align a dataset's needs with tailored coding tasks, such as data cleaning or transformation.

LLMs perform generally very well on popular datasets seen during training. For example, the prompt "generating an ML pipeline for the Titanic dataset" with GPT-4 creates an efficient ML pipeline. However, LLMs struggle with unknown datasets not seen during training, and generate errors due to hallucinations. Some studies have explored using LLMs for model selection and pre-processing in the form of feature engineering. Moreover, these approaches often face significant performance challenges when applied to large datasets with numerous features, which leads to performance degradation. Table 1 qualitatively compares state-of-the-art AutoML systems, such as FLAML [7] and Auto-Sklearn [2], as well as LLM-based systems, such as CAAFE [4], AIDE [6] and AutoGen [8].